Manuscript submission

Journal of Graph Theory 26 October 2026

Dear Editors,

Please consider the manuscript *Every odd antihole is strongly chromatic-choosable* by Deep Bhattacharjee and Ushashi Bhattacharya for publication in *Journal of Graph Theory*.

The manuscript proves that every odd antihole $\overline{C_{2\ell-1}}$, $\ell \geq 3$, is strongly ℓ-chromatic-choosable: its only non-colourable assignments of lists of size $\ell - 1$ are the constant assignments. It also gives the resulting extremal characterization in odd-hole-free critical graphs. These are Conjectures 9 and 3 of Fiujlaali and McDonald, whose contribution and deduction are credited. The proof proceeds through complements of odd linear forests, Hall's theorem, and a lemma on minimal tight sets. The supplied Lean sources formalise the main theorem; the critical-graph deduction is proved in the text. The paper fits the journal's interest in structural graph theory and list colouring.

An earlier version is publicly available as Hexagon 2610.00319v1. The source, verification code and supporting material are archived with the accompanying release:

`https://github.com/creelie/odd-antihole/releases/tag/v1.2.0`

The submission includes a separate declaration describing AI assistance and the scope of the formal verification.

Thank you for considering the manuscript.

Sincerely,

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